

Rational Löwner Approximation of Positive Monomials

Automated Math Discovery Candidate Proof

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Status. Candidate full solution, likely valid pending expert review. We prove a stronger multivariable theorem: every positive monomial of total exponent at most one is a compact-open limit of rational globally operator-monotone functions.

Source Question

Agler, McCarthy, and Young [1] prove that $(z^1 z^2)^s$ is globally operator monotone on $(0, \infty)^2$ for $0 \leq s \leq \frac{1}{2}$. Immediately after that example they ask:

We do not know if $(z^1 z^2)^s$ can be approximated by rational functions in the Löwner class.

The question occurs on source page 49 (PDF page 49) in the final section of arXiv:1009.3921v3, immediately after Example 9.8 begins on page 48; in the source archive this is `glob.tex`, lines 247–290.

if and only if $F \circ \rho_t$ is globally operator monotone on $\rho_t^{-1}(U) \cap (-\infty, 1/t)^d$.

(ii) The function F is globally operator monotone on $U \cap F^{-1}(-1/t, 1/t)$

if and only if $\rho_t \circ F$ is globally operator monotone on the same set.

What happens to Theorem 9.6 in $d \geq 3$ variables? It is still true that rational Löwner functions have finite-dimensional models [11, 8]. However, a recent example of Knese [22] shows that the minimal dimension n^r needed may be strictly greater than the degree of F in z^r . So we cannot rule out the possibility that the μ -spectrum of X contains some other algebraic sets in $\mathbb{R}^d$ than just the zero set of the denominator of F .

We now give an example of a non-rational globally operator monotone function.

Example 9.8. For $0 \leq s \leq 1/2$, the function $(z^1 z^2)^s$ is operator monotone on $(0, \infty) \times (0, \infty)$.

Indeed, if $(0, 0) < (A^1, A^2) \leq (B^1, B^2)$ and s is between 0 and $1/2$, then

$$\|(A^r)^s (B^r)^{-s}\| \leq 1 \quad \text{for } r = 1, 2.$$

Therefore the norm of

$$(B^1)^{-s}(A^1)^s(A^2)^s(B^2)^{-s}$$

is less than or equal to 1, so the largest eigenvalue is less than or equal to 1, and therefore the largest eigenvalue of

$$(B^2)^{-s/2}(B^1)^{-s/2}(A^1)^s(A^2)^s(B^1)^{-s/2}(B^2)^{-s/2} \quad (9.9)$$

is also less than or equal to 1. But (9.9) is self-adjoint, so less than or equal to the identity. Therefore

$$(A^1 A^2)^s \leq (B^1 B^2)^s.$$

We do not know if $(z^1 z^2)^s$ can be approximated by rational functions in the Löwner class.

We use the standard compact-open meaning of approximation. Thus a sequence on $(0, \infty)^d$ approximates a function if it converges uniformly on every compact subset.

Definitions

Let $E \subset \mathbb{R}^d$ be open. A real function f on E is *globally operator monotone* if

$$S^r \leq T^r \quad (1 \leq r \leq d) \quad \implies \quad f(S) \leq f(T)$$

whenever S and T are commuting d -tuples of bounded self-adjoint operators with joint spectra in E . The coordinates of S commute with one another, as do those of T ; no commutation between the two tuples is assumed.

The Löwner class $\mathcal{L}(E)$ is the boundary-value class defined in [1]. We need only their Theorem 8.1: for real C^1 functions on E , membership in $\mathcal{L}(E)$ is equivalent to local operator monotonicity. In particular,

$$\text{globally operator monotone on } E \implies f \in \mathcal{L}(E). \quad (1)$$

Proof Intuition

The source function looks nonrational because it contains fractional powers. The useful observation is that fractional powers and weighted geometric means are positive averages of very simple rational order-preserving maps.

For $0 < \theta < 1$, the scalar weighted geometric mean has the integral

$$x^{1-\theta}y^\theta = \frac{\sin(\pi\theta)}{\pi} \int_0^\infty \lambda^{\theta-1} (x^{-1} + \lambda y^{-1})^{-1} d\lambda.$$

Each integrand is jointly operator monotone: inversion reverses order, the positive sum preserves it, and the final inversion reverses it again. Similarly, t^p is a positive integral of the rational operator-monotone maps $t/(t+\lambda)$. Finite positive quadrature sums therefore give rational operator-monotone approximants.

Nested composition causes no loss. Functions of a fixed commuting tuple commute, and a globally operator-monotone binary map can be applied to any ordered pair of such outputs. This lets us build an arbitrary weighted geometric monomial recursively and then apply the outer power. The same positive integral formulas give uniform convergence on compact sets.

Main Theorem

Theorem 1. *Let $d \geq 1$ and let $a_1, \dots, a_d \geq 0$ satisfy*

$$p := \sum_{j=1}^d a_j \leq 1.$$

There is a sequence of rational functions R_n on $\mathbb{R}^d$, with no poles on $(0, \infty)^d$, such that:

- 1. each R_n is globally operator monotone on $(0, \infty)^d$;*
- 2. each R_n belongs to the Löwner class $\mathcal{L}((0, \infty)^d)$;*
- 3. $R_n \rightarrow \prod_{j=1}^d x_j^{a_j}$ uniformly on every compact subset of $(0, \infty)^d$.*

Corollary 1. *For every $0 \leq s \leq \frac{1}{2}$, the function $(z^1 z^2)^s$ on $(0, \infty)^2$ is a compact-open limit of rational functions in $\mathcal{L}((0, \infty)^2)$. Hence the question in [1] has an affirmative answer.*

Rational Building Blocks

Lemma 1 (Positive quadrature for a power). *For $0 < p < 1$, the function t^p on $(0, \infty)$ is a compact-open limit of finite positive sums of the rational functions*

$$q_\lambda(t) = \frac{t}{t + \lambda}, \quad \lambda > 0.$$

Every such finite positive sum is operator monotone and strictly positive on $(0, \infty)$.

Proof. Euler's beta integral gives

$$t^p = \frac{\sin(\pi p)}{\pi} \int_0^\infty \lambda^{p-1} \frac{t}{t + \lambda} d\lambda. \quad (2)$$

Indeed, the substitution $\lambda = tu$ reduces the integral to $t^p \int_0^\infty u^{p-1} (1 + u)^{-1} du$.

The map q_λ is operator monotone because

$$q_\lambda(A) = I - \lambda(A + \lambda I)^{-1},$$

and inversion reverses the order of positive invertible operators. Hence finite positive sums of the q_λ are operator monotone.

For $t \in [c, C] \subset (0, \infty)$, the part of (2) over $(0, \delta)$ is at most δ^p/p , uniformly in t , and the part over (M, ∞) is at most $CM^{p-1}/(1-p)$. On $[\delta, M]$ ordinary positive Riemann sums converge uniformly for $t \in [c, C]$. Letting $\delta \downarrow 0$, $M \uparrow \infty$, and the mesh tend to zero produces a single sequence converging on every compact subinterval of $(0, \infty)$. $\square$

Lemma 2 (Positive quadrature for a weighted geometric mean). *Let $0 < \theta < 1$. The function*

$$g_\theta(x, y) = x^{1-\theta} y^\theta$$

on $(0, \infty)^2$ is a compact-open limit of finite positive sums of the rational functions

$$h_\lambda(x, y) = (x^{-1} + \lambda y^{-1})^{-1} = \frac{xy}{y + \lambda x}, \quad \lambda > 0.$$

Every such sum is globally operator monotone and strictly positive on $(0, \infty)^2$.

Proof. The beta integral and the substitution $\lambda = (y/x)u$ give

$$x^{1-\theta}y^\theta = \frac{\sin(\pi\theta)}{\pi} \int_0^\infty \lambda^{\theta-1} \frac{xy}{y+\lambda x} d\lambda. \quad (3)$$

Suppose $A_1 \leq B_1$ and $A_2 \leq B_2$ are positive invertible operators. Then

$$A_1^{-1} + \lambda A_2^{-1} \geq B_1^{-1} + \lambda B_2^{-1},$$

so another inversion yields

$$(A_1^{-1} + \lambda A_2^{-1})^{-1} \leq (B_1^{-1} + \lambda B_2^{-1})^{-1}.$$

When each pair is commuting, these are exactly the joint functional-calculus values of h_λ . Thus h_λ is globally operator monotone.

For $(x, y) \in [c, C]^2$, the integrand without the constant in (3) is bounded by $C\lambda^{\theta-1}$ near zero and by $C\lambda^{\theta-2}$ near infinity. The tails therefore tend uniformly to zero. Positive Riemann sums on a truncated compact λ -interval give the asserted rational approximants and compact-open convergence. $\square$

Composition and Proof of the Theorem

Lemma 3 (Order-preserving composition). *Let U, V be strictly positive globally operator-monotone functions on $(0, \infty)^d$, and let H be globally operator monotone on $(0, \infty)^2$. Then*

$$x \mapsto H(U(x), V(x))$$

is globally operator monotone on $(0, \infty)^d$. If all three functions are rational and have no poles on their positive domains, then so is the composition.

Proof. For ordered commuting d -tuples $S \leq T$, global monotonicity gives $U(S) \leq U(T)$ and $V(S) \leq V(T)$. The pair $(U(S), V(S))$ commutes because both entries belong to the joint functional calculus of S ; the analogous pair for T also commutes. The defining property of H now gives

$$H(U(S), V(S)) \leq H(U(T), V(T)).$$

The rationality assertion is immediate from substitution, and positivity keeps all positive-domain denominators away from zero. $\square$

Proof of Theorem 1. The case $p = 0$ is the constant function 1. Coordinates with $a_j = 0$ may be omitted, so assume first that the active exponents are positive. Put

$$w_j = \frac{a_j}{p}, \quad \sum_j w_j = 1.$$

We first approximate the normalized weighted monomial $G(x) = \prod_j x_j^{w_j}$.

Order the active coordinates and set $W_k = \sum_{j=1}^k w_j$. Recursively define

$$G_1(x) = x_1, \quad G_k(x) = g_{w_k/W_k}(G_{k-1}(x), x_k) \quad (k \geq 2).$$

A scalar exponent calculation gives

$$G_k(x) = \prod_{j=1}^k x_j^{w_j/W_k};$$

hence $G_d = G$.

For every binary g_{w_k/W_k} choose the rational positive quadrature approximants from Lemma 2, and substitute them recursively. Lemma 3 shows that every resulting nested rational function is globally operator monotone. A standard diagonal choice makes these nested functions converge to G uniformly on compact subsets. For completeness, the induction is harmless: on any fixed compact domain, G_{k-1} has a compact range bounded away from zero; uniform convergence of the preceding stage puts all sufficiently late approximating ranges in a slightly larger compact subinterval of $(0, \infty)$, where the next binary quadrature converges uniformly.

If $p = 1$, these nested approximants already converge to $G = \prod_j x_j^{a_j}$. If $0 < p < 1$, compose them with the positive rational operator-monotone approximants to t^p from Lemma 1. Another application of Lemma 3, followed by the same compact-range diagonal argument, yields rational globally operator-monotone functions converging compact-open to

$$G(x)^p = \prod_{j=1}^d x_j^{a_j}.$$

Finally, every approximant belongs to $\mathcal{L}((0, \infty)^d)$ by (1). This proves all three assertions. $\square$

Proof of Corollary 1. Take $d = 2$ and $a_1 = a_2 = s$. The hypothesis of Theorem 1 is exactly $2s \leq 1$. The endpoint $s = 0$ is the constant 1, so the entire range in the source question is covered. $\square$

Verification and Limitations

No numerical or symbolic computation is used. The proof reduces to the two positive integral formulas (2) and (3), two order reversals under inversion, and a composition argument inside joint functional calculus.

The packet proves locally uniform approximation on $(0, \infty)^d$. It does not claim uniform approximation up to the boundary, where the monomials may lose smoothness, or a degree-optimal approximation rate.

A bounded search on August 9, 2026 used the exact source phrase and close variants involving rational Löwner approximation, operator-monotone monomials, and weighted geometric/harmonic means. It found the source paper and later one-variable approximation work, but no explicit answer to the question or statement of Theorem 1. Since all ingredients are classical, novelty confidence is moderate-to-low pending specialist review.

References

- [1] J. Agler, J. E. McCarthy, and N. J. Young, *Operator monotone functions and Löwner functions of several variables*, Ann. of Math. (2) **176** (2012), 1783–1826; arXiv:1009.3921.